

Eshopbox: Enabling brands to fulfill next-day delivery with App Engine

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Eshopbox is an all-in-one ecommerce

Eshopbox's scalable platform on App allows brands to ecommerce operations gain insight into demand through while optimizing through BigQuery



About Eshopbox

Eshopbox is an all-in-one ecommerce logistics platform using modern software to provide fast and affordable fulfillment. The company helps brands manage their ecommerce lifecycle from inventory management and order fulfillment to delivery and returns, with its digital platform. For quick delivery at affordable rates, Eshopbox stores inventory on behalf of its customers in fulfillment centers across India and works with multiple courier partners.

Industries: Technology, Retail & Consumer Goods

Google Cloud results

- Real-time integration with external platforms using Pub/Sub instead of running hourly scripts
- Improves delivery speed by analyzing shipping data with BigQuery
- Builds a no-code platform using Cloud Run to automate workflows in fulfilment centers in minutes

Location: India

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Engine (<https://cloud.google.com/appengine/>),

Google Cloud (<https://cloud.google.com/>),

Cloud Storage

BigQuery (<https://cloud.google.com/bigquery/>)

SQL (<https://cloud.google.com/sql/>) Cloud

App Engine (<https://cloud.google.com/appengine/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud SQL (<https://cloud.google.com/sql/>) help.

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

- Scales
effortlessly
to match
spikes in
customer
demand on
App Engine

There has been a sharp increase in ecommerce (<https://news.un.org/en/story/2021/05/1091182>) in recent years as more people discover the convenience of shopping anytime, anywhere, with the added benefit of having their goods delivered to their doorstep. Consumers also expect their goods to arrive quickly, and with so much competition in the online retail space, this has put pressure on retailers to keep enough stock on hand for next-day delivery.

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With Google
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improving our
platform for
retailers without
worrying about
server
optimization.”

*—Mayur Karwa, co-
founder and Chief
Technology Officer,
Eshopbox*

Eshopbox (<https://www.eshopbox.com/>), a third-party logistics company, helps brands scale their ecommerce business by outsourcing inventory management and order fulfilment. Powered by **Google Cloud** (<https://cloud.google.com/>), Eshopbox offers a fulfilment-as-a-service platform so brands can focus on selling online instead of operations. Using the Eshopbox workspace, brands can sell their products across multiple sales channels and track stock, orders, and payments in real time without having to manage the infrastructure on their own. The company stores inventory in its growing network of fulfilment centers across India and works

with courier partners to achieve fast and affordable delivery.

“We help brands and retailers eliminate the need to maintain physical warehouses, just like Google Cloud eliminates the need for physical servers,” says Mayur Karwa, co-founder and Chief Technology Officer at Eshopbox. “With Google Cloud, we can focus on improving our platform for retailers without worrying about server optimization.”

Eshopbox hosts its ecommerce management platform on [App Engine](#) (/appengine) for developer agility. For real-time insights, Eshopbox uses a combination of [Pub/Sub](#) (/pubsub) for data delivery to its primary data storage, [Cloud Storage](#) (/storage), and [BigQuery](#) (/bigquery) and [Cloud SQL](#) (/sql) for data analysis.

“Each team
takes ownership
of their own App
Engine
instances and
monitors costs
and usage,
without a
centralized

DevOps team as
the bottleneck.
App Engine
gives us the
agility to release
a new feature
every two weeks,
three times
faster than
physical
servers.”

*—Mayur Karwa, co-
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Enabling innovation on microservices architecture with App Engine

When Eshopbox joined [Google Cloud for Startups](#) (/startup) in 2016, the team took advantage of cloud credits to explore Google Cloud products. At that point, Eshopbox needed a fully managed environment to free up its lean team of five developers from operational tasks.

Fast forward to 2021, Eshopbox now runs a team of 50 developers, who build different features for Eshopbox workspace with the help of App Engine.

Using a microservices architecture on App Engine, each project team can develop, test, deploy, and scale their services independently from other services to speed up the software release cycle.

“Each team takes ownership of their own App Engine instances and monitors costs and usage, without a centralized DevOps team as the bottleneck,” says Karwa. “App Engine gives us the agility to release a new feature every two weeks, three times faster than physical servers.”

Project teams optimize cloud costs by selecting the appropriate scaling configuration, from manual, basic, and automated, that’s available on App Engine to match their use case. For example, engineers can choose the auto-scaling option to scale up instances during holiday seasons and flash sales when more people are shopping online and scale back down to a minimum number of instances when demand is lower.

“Ecommerce is a fast-moving business. App Engine gives us a scalable system that keeps up with rapid changes in sales volume for a smooth user experience,” says Karwa. “On top of that, the team only pays for what they use. This transparent pricing approach helps us optimize resources as we continue to build the business.”

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—Mayur Karwa, co-
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Updating inventory in real time with Pub/Sub

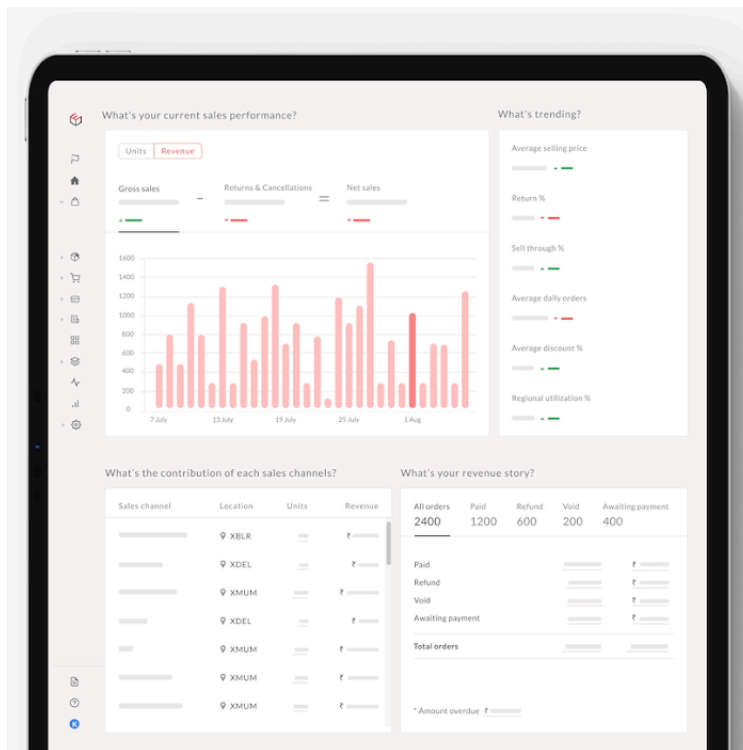
In the digital world where sales happen around the clock, retailers need to update inventory in real time to avoid situations where shoppers place an order on an item that has run out of stock. Previously, a developer ran a script every hour to check for event changes, updating each sales channel manually.

This laborious process often led to communication delays that affected customer experience. Now, Eshopbox improves the communication between teams and third-party systems by using Pub/Sub to capture data changes from different sources on Cloud Storage and relies on Cloud SQL to run queries. For example, if there's a shipping delay, Eshopbox automatically sends a notice to inform the customer so that their expectations are managed.

"We work with many stakeholders in our ecosystem, from online marketplaces to courier partners, so service-to-service communication is critical for a smooth user experience," says Karwa. "Pub/Sub connects our services to each other and improves our real-time integration with third-party systems without manual intervention."

This tight integration enables sellers to track product availability across sales channels on

Eshopbox workspace and stock up appropriately for each location to meet customer demand.



Automate workflows in warehouses within minutes using Cloud Run

To stay relevant in such a fast-paced environment like online retail, companies must be able to react nimbly and respond swiftly. Legacy warehouse management software is often unable to provide the flexible functionality required and has long implementation periods. Coupled with the fact that the average ecommerce business has more than 129 apps deployed across the organization, it is critical to look for a solution that supports integration with existing tools. For the flexibility it required, the team at Eshopbox knew they needed to

build a no-code solution and automate workflows by deploying them on Cloud Run.

“Our no-code automation framework allows clients to easily navigate workflows with dependencies on other systems without having to make any changes to their tech stack,” says Karwa. Using a platform that’s easily compatible is one big hurdle out of the way for Eshopbox’s customers.

Cloud Run also enables the team to keep an eye on important productivity metrics for operators at the fulfilment centers, and can periodically reassess workflows and optimize its processes every minute. This helps the team work smarter over time. To illustrate, the information on Cloud Run allows the team to create customized workflows based on categories like fast-moving consumer goods, food and nutrition, health and beauty, electronics, and fashion, the team can cater for organizations of all sizes, from startups to publicly listed conglomerates.

Improving on-time delivery with smart analytics on BigQuery

For on-time delivery, Eshopbox integrates 88 gigabytes of shipment data daily from carriers via API and analyzes shipping trends with BigQuery. By looking at data points such as geographic location, on-time delivery percentage, and delivery speed, Eshopbox can choose the right courier partner for each store-keeping unit (SKU) or product code by the postal index number (PIN) code for the delivery location.

“Every courier partner has their strengths and weaknesses,” says Karwa. “BigQuery helps us derive intelligence from a global level. By analyzing parcel deliveries across all our courier networks, we can route orders to the most suitable courier partner for the fastest delivery, meeting customer expectations.”

He adds, “Ecommerce has seen a surge in the post-COVID world. We continue to build new features such as one-day or two-day delivery so sellers can compete effectively with global online retailers. With the help of Google Cloud, we provide the key to success for online retailers, and that’s seamless and fast order fulfilment to win customer loyalty.”

